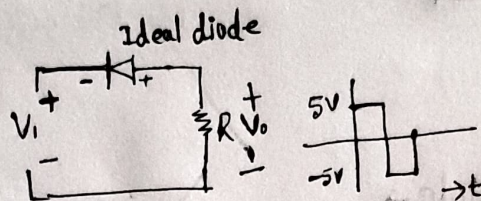


Clipper - Clipper are network that used to diode to clip a portion of input signal without distroying other.

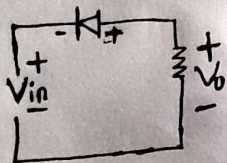
Usage $\rightarrow$ to clip noise portion from signal

①



Analysis $\Rightarrow$

for positive half-cycle (+5V) $\rightarrow$

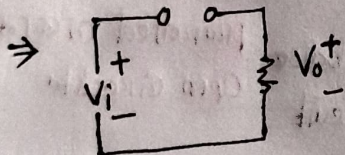


Plus প্রান্তের সাথে

Diode এর (-) প্রান্ত $\Rightarrow$ Reverse bias

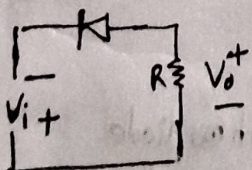
$\Rightarrow$ Open circuit

forward bias $\rightarrow$ short circuit
Apply KVL
Reverse bias $\rightarrow$ Open circuit
No current



$$V_o = 0$$

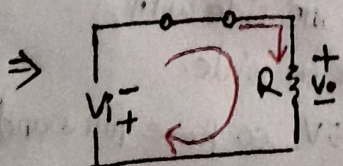
for negative half cycle (-5V) $\rightarrow$



(-) প্রান্তের সাথে

Diode এর (-) প্রান্ত $\Rightarrow$ Reverse forward bias

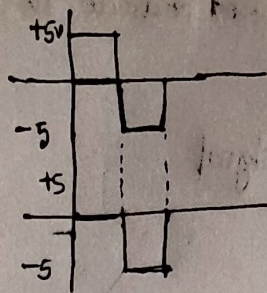
$\Rightarrow$ ~~Open circuit~~ short circuit



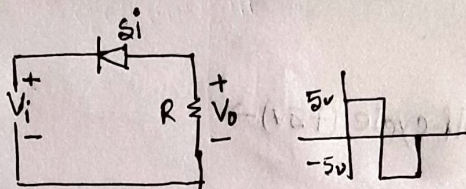
$$KVL = +V_o + V_i = 0$$

$$\therefore V_o = -V_i \Rightarrow V_o = -5$$

Output signal $\rightarrow$

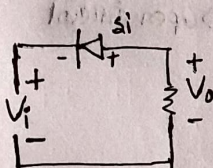


② Same circuit with si diode $\rightarrow$



Analysis $\Rightarrow$

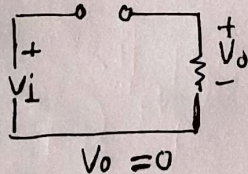
for positive half cycle (+5V) $\rightarrow$



(+) side with (-) side $\Rightarrow$ Reverse bias $\Rightarrow$ Open circuit

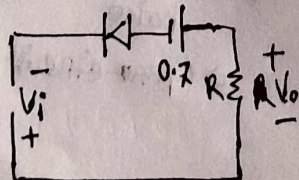
Reverse bias is ideal / practical মেকেন হাতে Open circuit.

$\Rightarrow$



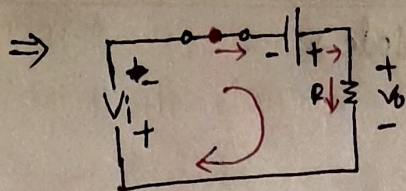
Practical diode is backward voltage প্রাপ্ত এবং এর চিহ্ন diode এর অনুসরণ করে.

for negative half cycle $\rightarrow$



(-) side with (+) side. কিন্তু diode এর 0.7 এর (-) side with diode এর (+) side.

But $0.7 < 5V$ so forward bias $\Rightarrow$ short circuit

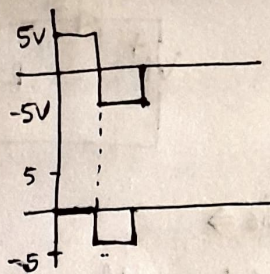


$$KVL = -0.7 + V_o + V_i$$

$$V_o = -5 + 0.7$$

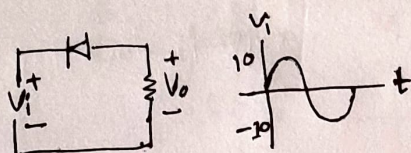
$$= -4.3$$

Output signal →



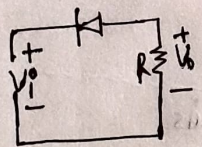
~~Output sign~~

III same circuit with ~~analog~~ Analog wave -

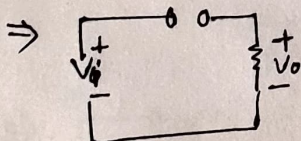


Analysis ⇒

for positive half cycle (+10) →

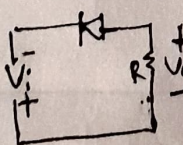


⇒ (+) side with (-) side
 ⇒ Reverse bias
 ⇒ Open circuit

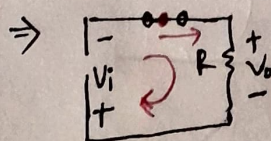


$$V_o = 0$$

for negative half cycle (-10) →

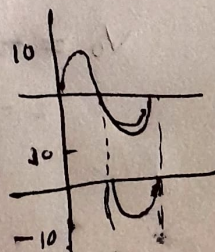


⇒ (-) side with (-) side
 ⇒ forward bias
 ⇒ short circuit

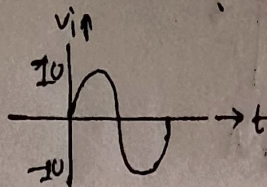
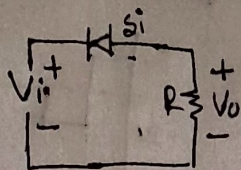


$$KVL = +V_o + V_i \therefore V_o = -V_i$$

Output signal →

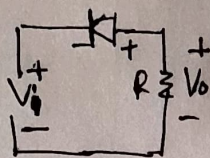


④ Same circuit, same wave with practical diode $\rightarrow$



Analysis $\Rightarrow$

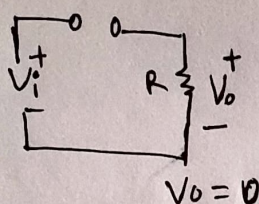
for positive half cycle ($+10$) $\rightarrow$



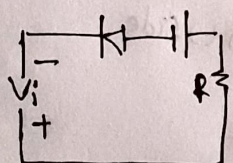
$\Rightarrow$ Reverse bias

$\Rightarrow$ open circuit

$\Rightarrow$



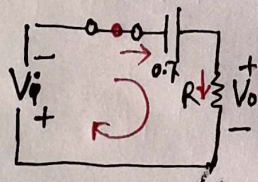
for negative half cycle $\rightarrow$



$\Rightarrow$ forward bias

$\Rightarrow$ short circuit

$\Rightarrow$



for $V_i < 0.7V$, $V_o = 0$

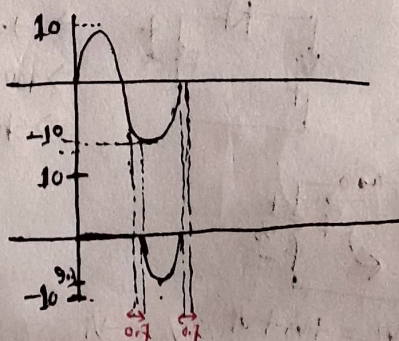
for $V_i > 0.7V$,

$$KVL = -0.7 + V_o + V_i$$

$$V_o = -V_i + 0.7$$

$$= -9.3V$$

Output signal $\rightarrow$



গাণিত্য →

Analogy wave form এ Voltage মান constant আছে না।

Practical diode এর জন্য circuit এ ~~কিছু~~ Backward voltage থাকবে 0.7V.

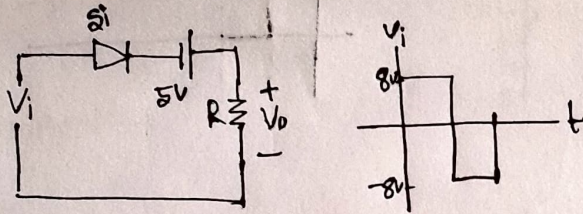
সি Backward voltage এর জন্য diode reverse bias এ আছে। কাজে

Analogy wave 0.7 মানের circuit 0.7 কে প্রবর্তন দিয়ে reverse

bias এ থাকবে এবং voltage output voltage = 0 হবে। তারপর বাকি অংশ

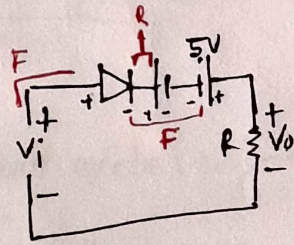
forward bias Apply হবে।

(V)

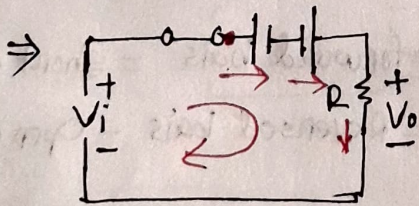


Analysis →

for positive half cycle (+8V) →



forward bias
⇒ short circuit



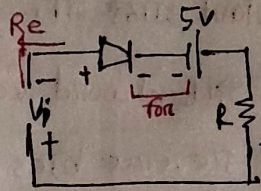
$$KVL = 0.7V + 5V + V_o - 8 = 0$$

$$\Rightarrow V_o = -0.7 + 5 + 8$$

$$= 13 - 0.7$$

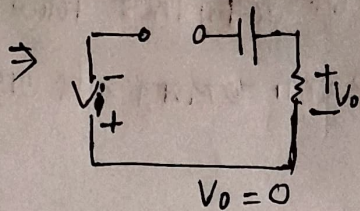
$$= 12.3$$

for negative half cycle $(-8) \Rightarrow$

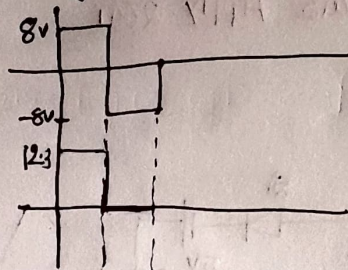


As $5V < 8V$

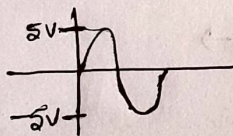
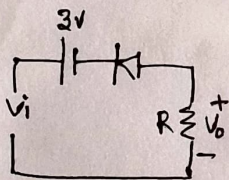
$\therefore$ Reversed biased
 $\Rightarrow$ Open circuit



Output signal $\rightarrow$

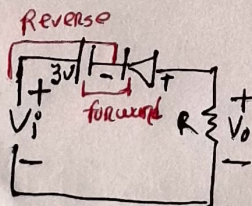


(vi)



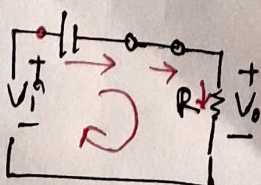
Analysis $\Rightarrow$

for positive half cycle $(+5V) \Rightarrow$



for $V_i < 3V =$ forward bias = short circuit
 $V_i > 3V =$ Reversed bias = Open circuit

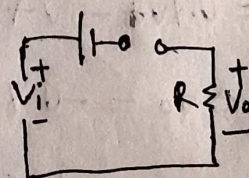
$V_i < 3V \Rightarrow$



$$KVL = 3 + V_o - V_i = 0$$

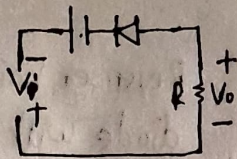
$$\therefore V_o = -3 + V_i = 2V$$

$V_i > 3V \Rightarrow$

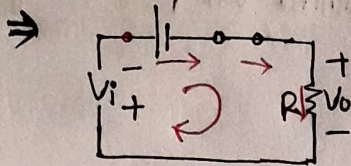


$$V_o = 0$$

for negative half-cycle $(-8.5V) \rightarrow$



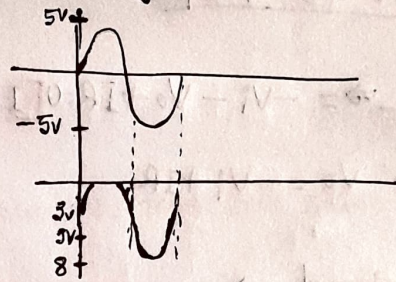
forward bias for both $3V$ and input voltage
= short circuit



$$KVL = 3 + V_o + 5$$

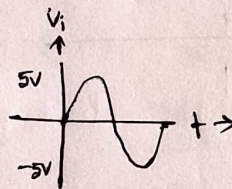
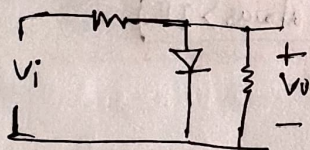
$$V_o = -3 - 5 \\ = -8$$

output signal $\rightarrow$



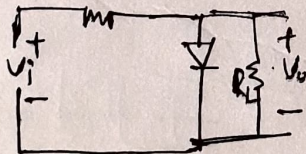
complicated circuit $\rightarrow$

①



Analysis $\Rightarrow$

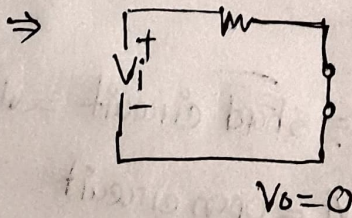
for positive half cycle $(+5V) \rightarrow$



forward biased
diode will be short circuit

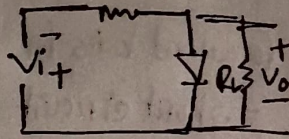
$$R_L = 0 \therefore V_o = 0$$

RL याकालेन छान्ने diode ला short
circuit छान्ने diode फिर्त
pass छान्ने Load ला 0 छान्ने

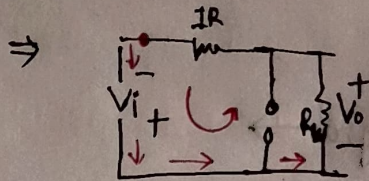


$$V_o = 0$$

for negative half cycle $(-5) \rightarrow$

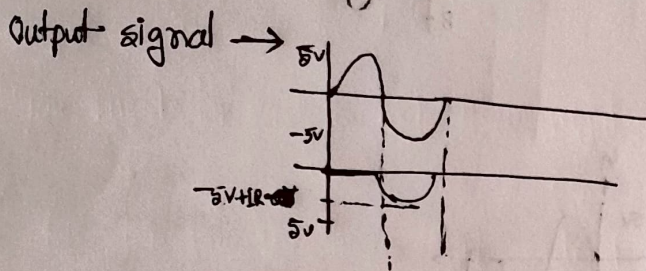


Reverse biased
diode will be open circuit
All current will pass through R_L



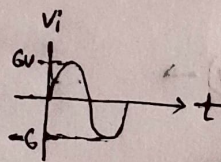
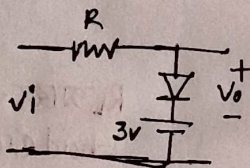
KVL $= -V_i - V_o + IR = 0$ [I Always + ফিকে দ্বারা করতে]

$\therefore V_o = -V_i + IR$



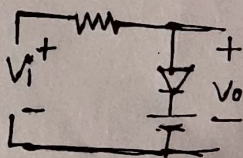
$V_i \text{ always} > IR$

(ii)



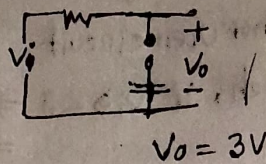
Analysis $\Rightarrow$

for positive half cycle $(+6V) \rightarrow$

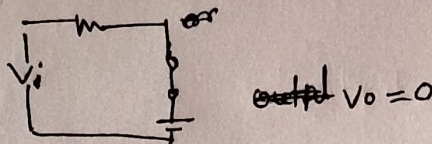


forward bias for $3V =$ short circuit $=$ output $= 0$
Reverse bias for $V_i =$ open circuit

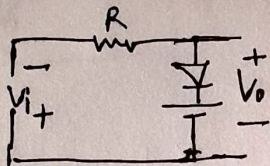
When $V_i < 3V \rightarrow$



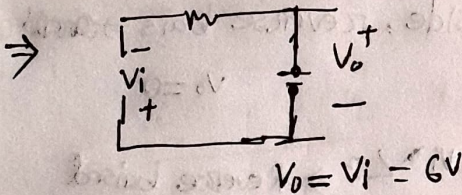
When $V_i > V_o \rightarrow$



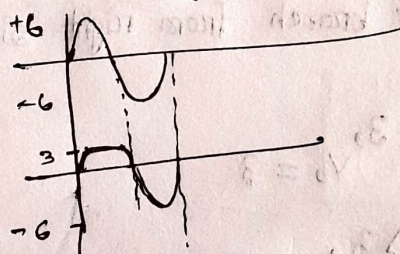
for positive half cycle ($-6V$) $\rightarrow$



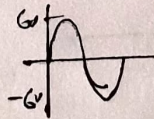
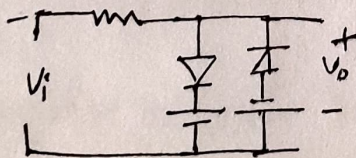
Reverse bias from both side



output signal $\rightarrow$

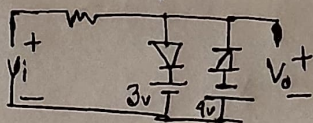


(III)



Analysis $\Rightarrow$

for positive half cycle ($+6V$) $\rightarrow$



for 1st branch from right, there is reverse bias $\Rightarrow$ short open circuit

for 2nd branch, if $V_i < 3 \Rightarrow$ Reverse bias $\Rightarrow$ open circuit

for $V_i < 3$,

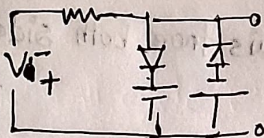
$$V_o = 3V$$

for $V_i > 3$,

$$V_o = 0$$

$V_i > 3 \Rightarrow$ forward bias $\Rightarrow$ short circuit

for negative half cycle $(-6V) \rightarrow$



for first or 2nd branch from right side, reverse bias $\Rightarrow$ short open circuit

$$V_o = 0$$

for 1st branch from right side if $V_i < 3 \Rightarrow$ Reverse biased

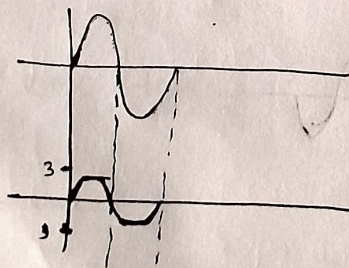
for $V_i < 3$,

$$V_o = 3$$

for $V_i > 3$,

$$V_o = 3$$

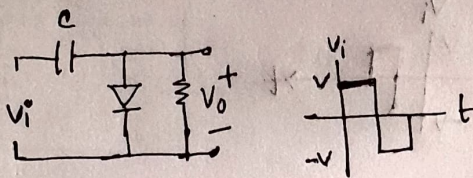
output signal $\rightarrow$



Clamper $\rightarrow$

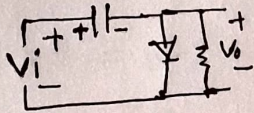
A clamper is a network constructed of a diode, resistor and capacitor that shift the waveform to a different level without changing the appearance of the applied signal.

①



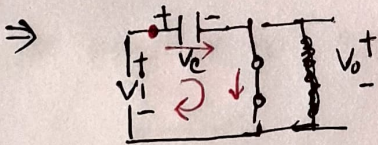
Analysis $\Rightarrow$

for positive half cycle $(+V) \rightarrow$



forward biased
 $\Rightarrow$ short circuit

capacitor charge will be V_i এর similar হবে।

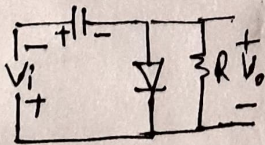


$$V_o = 0$$

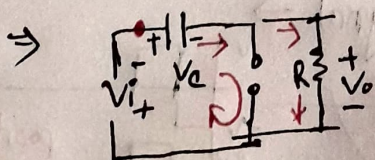
Using KVL we get, $+V_c - V_o = 0$

$\Rightarrow V_c = V$ (capacitor charging rate)

for negative half cycle $(-V) \rightarrow$



Reverse biased
 $\Rightarrow$ open circuit

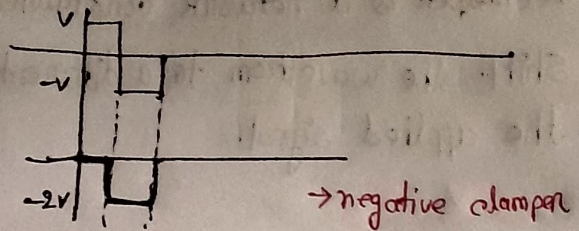


$$\text{KVL} \Rightarrow +V_c + V_o + V_i = 0$$

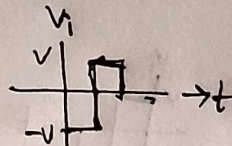
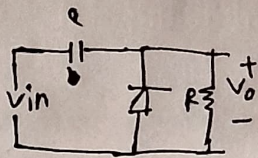
$$V_o = -V_c - V$$

$$= -V_c - 2V$$

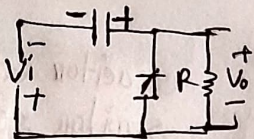
Output signal $\rightarrow$



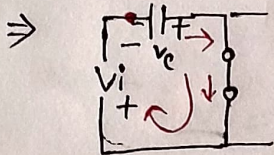
①



for negative half cycle $(-V) \rightarrow$



forward biased
 $\Rightarrow$ Short circuit

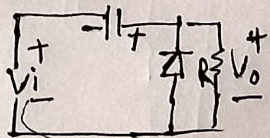


$$V_o = 0$$

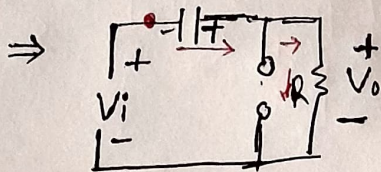
Using KVL we get, $-V_c + V_i = 0$

$$V_c = V_i$$

for positive half cycle $(+V) \rightarrow$



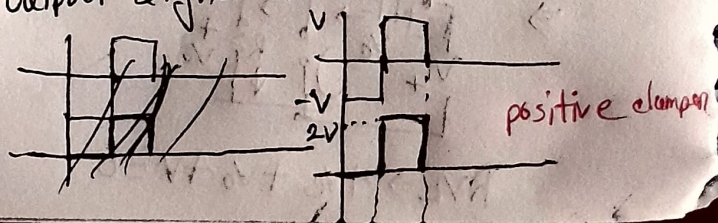
Reverse biased
 $\Rightarrow$ Open circuit



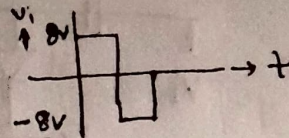
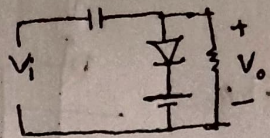
$$KVL = -V_c + V_o - V = 0$$

$$V_o = 2V$$

output signal $\rightarrow$

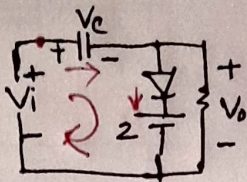


(iii)



Analysis $\Rightarrow$

for positive half-cycle (+8V) $\rightarrow$



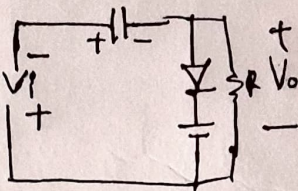
forward biased
 $\Rightarrow$ short circuit

$V_o = 2V$ [diode forward voltage drop during forward bias]

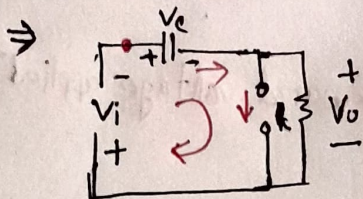
$$KVL \Rightarrow V_c + 2 - 8 = 0$$

$$V_c = 8 - 2 \\ = 6$$

for negative half cycle (-8V) $\rightarrow$



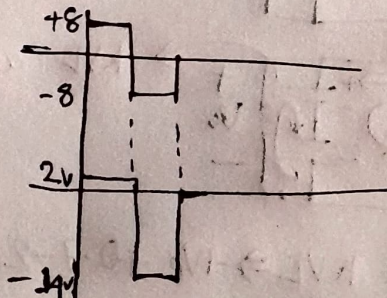
open circuit
Reverse biased
 $\Rightarrow$ Open circuit



output v_o signal $\Rightarrow$

$$\text{Using KVL} \Rightarrow +V_c + V_o + V_i$$

$$\Rightarrow V_o = -6 - 8 \\ = -14$$

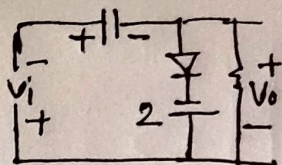


~~V_o = -1.3~~

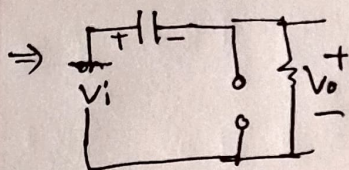
$$-V_o + 0.7 - 2 = 0$$

$$V_o = -1.3$$

for negative half cycle →



Reverse biased
⇒ ~~Short~~ Open circuit



$$\text{KVL} \Rightarrow +V_i + V_o + 10 = 0$$

$$V_o - 11.3 + V_o + 10 = 0$$

$$\Rightarrow V_o = -21.3$$

Output signal →

